Gaussian 16 NMR Simulator

Student Manual

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Introduction

This tool converts raw Gaussian 16 quantum chemistry output files, such as `.out` or `.log` files, into annotated one-dimensional ^{1}H NMR spectra.

The simulator extracts magnetic shielding tensors, computes chemical shifts relative to a chosen reference compound such as TMS, groups equivalent protons, applies J-coupling splitting rules when spin-spin data are available, and generates a visual spectrum suitable for interpretation and discussion.

1 Prerequisites and Installation

To run the simulator, install **Python 3** and the two required scientific libraries: `numpy` and `matplotlib`. From a command prompt or terminal, run:

```
pip install numpy matplotlib
```

Download the simulator script, `nmr_spectrum_simulator.py`, and place it in your working directory together with the relevant Gaussian output files.

2 Preparing Your Gaussian 16 Files

The simulator requires **two** separate `.out` or `.log` files:

1. **Reference file**: usually tetramethylsilane, TMS.
2. **Target molecule file**: the molecule whose NMR spectrum you want to simulate.

2.1 Running the Reference Job

Compute the isotropic shielding of the reference compound at the **same level of theory and basis set** used for the target molecule. This consistency is essential because chemical shifts are obtained by comparing the target shieldings against the reference shielding.

Example route section:

```
# B3LYP/6-311G** Opt Freq NMR=giao
```

2.2 Running the Target Job

Optimize the target molecule and request NMR properties. The route section depends on whether you want only chemical shifts or also multiplet splitting:

- **Singlet-only spectrum:** use `NMR=giao`.
- **Multiplet splitting with J-couplings:** explicitly request spin-spin couplings using `NMR=SpinSpin`.

Example route sections:

```
# B3LYP/6-311G** Opt Freq NMR
# B3LYP/6-311G** Opt Freq NMR=SpinSpin
```

Warning

If the Gaussian input file omits the `NMR` keyword, the script will not find the `Magnetic shielding tensor` block in the output file. In that case, the calculation must be rerun with the correct NMR keyword before the simulator can process the file.

3 How to Run the Simulator

You can run the script either from the command line or through interactive prompts.

3.1 Method A: Command Line

Use the following syntax:

```
python nmr_spectrum_simulator.py <path_to_target.out> <path_to_tms.out>
```

Example:

```
python nmr_spectrum_simulator.py aceto1.out tms1.out
```

3.2 Method B: Interactive Prompts

If you run the script without arguments, the program asks you to type the paths to the two files:

```
python nmr_spectrum_simulator.py
```

3.3 Changing the Magnetic Field

By default, the simulator mimics a **400 MHz** NMR spectrometer. To simulate another field strength, use the `-f` or `-freq` flag. For example, to simulate a 600 MHz spectrometer:

```
python nmr_spectrum_simulator.py aceto1.out tms1.out -f 600
```

Increasing the spectrometer frequency narrows the apparent separation of multiplet lines on the ppm scale because J-couplings are measured in hertz, while the spectrum is plotted in ppm.

4 Understanding the Output

After the script finishes, it prints a summary of the extracted proton groups to the console and generates an image file named:

```
NMR_G16_<molecule_name>.png
```

4.1 The Printed Summary

The terminal output may look like this:

```
Computed TMS Reference Shielding: 31.9257 ppm
Found spin-spin couplings! Using a threshold of 0.5 Hz for splitting.
--- Processed Molecule ---
Name: aceto1
  delta 10.34 (td (3x2 lines), 1H) - Group A
  delta 2.16 (dd (2x2 lines), 2H) - Group B
  delta 1.92 (dt (2x3 lines), 1H) - Group C
```

A note on static geometry vs. real-world NMR

The case of the split methyl group. In acetaldehyde, you might expect the $-CH_3$ group to appear as a single 3H peak. However, a static quantum-chemical calculation may split it into a 2H group and a 1H group.
Standard Gaussian optimizations calculate properties for a **static, 0 K gas-phase molecule**. Without explicitly modelling molecular rotation or dynamics, the three methyl protons are fixed in space. One proton may lie in the plane of the carbonyl group while the other two are out of plane, producing slightly different magnetic environments. In a real room-temperature NMR experiment, methyl rotation is very fast. The spectrometer therefore observes a *time-averaged* chemical shift. This contrast between static quantum-mechanical structures and dynamic experimental conditions is an important concept in computational chemistry.

5 Checklist Before Running

Before launching the simulator, verify that:

- the reference and target calculations used the same method and basis set;
- both output files contain NMR shielding data;
- the target calculation includes `NMR=SpinSpin` if multiplet splitting is required;
- the file paths used in the command line are correct;
- `numpy` and `matplotlib` are installed in the Python environment you are using.